Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program
(TIPA: Korea Technology and Information Promotion Agency for SMEs)

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Energy-Efficient Edge AI for Smart Manufacturing and Advanced Nuclear Systems: Quantized and Neuromorphic Foundation Models			
Korea's 12 National Strategic Technology Fields	Next-gen Nuclear Power (√), AI (√), and Advanced Robotics & Manufacturing (√)			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication		Energy and Resources	
	Medium Category	Small Category	Medium Category	Small Category
	Software	Software Solutions	Nuclear Power	Next-Generation Nuclear Power
Concept of Proposed Technology	This project delivers a deployable foundation model for virtual sensing (FM-VS) in Smart Manufacturing and Advanced Nuclear Systems. Built on neural-operator architectures, the model learns a parametric solution operator that maps sparse and asynchronous sensor measurements to high-dimensional internal-state fields, enabling direct on-site inference under strict power and connectivity constraints and allowing industrial operators to reconstruct states that cannot be physically instrumented within manufacturing equipment or energy assets.			
	This capability resolves a structural barrier in industrial AI deployment. Critical components in advanced manufacturing lines, thermal processing systems, energy infrastructure, and next-generation nuclear facilities operate under extreme temperature, radiation, pressure, or geometric constraints where dense instrumentation is infeasible or cost-prohibitive. Internal states must therefore be inferred from limited external sensing, yet conventional physics-based solvers are computationally incompatible with real-time execution on low-power edge hardware. The result is a persistent monitoring gap between the operational visibility required for reliability, yield optimization, and regulatory compliance, and what can be computed in situ within SME-scale industrial environments.			
	The project closes this gap through three tightly integrated thrusts: 1. **Transferable operator architecture**. The FM-VS core approximates the underlying nonlinear PDE solution operator rather than isolated state trajectories, enabling systematic transfer across operating regimes, geometries, manufacturing configurations, and sensor layouts with minimal re-tuning. This reduces data burden, shortens integration timelines, and allows Korean SMEs to deploy a unified AI backbone across multiple product platforms or industrial assets. 2. **Deployment-grade validation.** Performance is quantified in field-relevant units, including energy per inference, tail latency, and robustness under sensing sparsity and drift. These metrics establish sustained edge feasibility and long-duration operational reliability, aligning AI performance with industrial power envelopes, uptime			

	requirements, and product-level integration constraints rather than relying solely on laboratory accuracy. 3. ***Hardware and software co-design.*** *The operator backbone is profiled and optimized across heterogeneous edge platforms, from GPU-class accelerators to quantized VPU-class and neuromorphic processors, enabling partners to align computational pathways with product-specific power budgets, embedded-system constraints, and continuous-operation requirements typical of smart factories and energy systems.* *By establishing energy-efficient and integration-ready virtual sensing for Smart Manufacturing and Advanced Nuclear Systems, FM-VS provides the perception substrate required for autonomous and real-time monitoring in safety-critical and productivity-critical environments. This capability transforms foundation models from laboratory demonstrations into deployable industrial infrastructure for next-generation manufacturing and energy systems.*
Rationale for Joint R&D Project	Foundation models have achieved sub-1 percent reconstruction error in controlled laboratory studies, yet none have transitioned into sustained industrial deployment across manufacturing and advanced energy systems. The barrier is not predictive accuracy. The barrier is the absence of hardware-validated deployment evidence under real industrial constraints. Industrial partners require quantified power budgets, bounded latency on representative edge hardware, and demonstrated robustness under sensor sparsity and drift before integration into production and safety-critical workflows. These requirements cannot be satisfied by academic prototypes alone. They require structured co-development that converts operator-learning research into field-validated, integration-ready industrial capability. *This collaboration directly addresses that conversion gap. The Korean SME contributes the defining elements of product realization: target asset class, sensing topology, interface constraints, deployment environment, and acceptance criteria tied to operational decision thresholds in smart factory and advanced energy contexts. UIUC provides the FM-VS operator core, quantization and neuromorphic optimization pathways, and a standardized benchmarking protocol that reports energy per inference, tail latency, and uncertainty coverage in deployment-relevant units. The integration of these roles produces not just a model, but a validated deployment framework that enables commercialization with measurable risk reduction and shortened integration timelines.* Through established collaborations with Idaho National Laboratory, Sandia National Laboratories, and Oak Ridge National Laboratory, the partnership gains access to domain-specific validation scenarios, representative datasets, and heterogeneous hardware testbeds that can inform the advancement of FM-VS from prototype to field-validated technology, including relevance to next-generation nuclear power systems. This expands the SME's technical validation envelope while accelerating time to deployable capability in advanced manufacturing and energy environments that would otherwise require multi-year coordination. The PI's national and international leadership [1,2,3,4,5] further reduces strategic and technical risk for the partnership. As an **Expert Committee Member of the National Academies of Sciences, Engineering, and Medicine (NASEM) on Foundation Models for Scientific Discovery and Innovation**, advising the U.S. Department of Energy on national strategy for foundation models, the project is directly aligned with federal scientific infrastructure priorities. Selection among **100 Global Leaders by the Simons Foundation** situates the effort within the international community shaping next-generation AI for science. Recognition as a **National AI Leader in the White House AI Action Plan response**, recipient of the **HPCwire Editors' Choice Award in Energy**, and **Illinois Innovation Award finalist for digital twin commercialization** demonstrates translational impact beyond academic research. Additional distinctions including **ANS 40 Under 40** and top-tier global research visibility reinforce established credibility. Collectively, these positions strengthen validation pathways, policy alignment, and long-term sustainability of the joint technology.
Keywords	1. Neuromorphic Foundation models. 2. Energy-Efficient Edge AI Deployment 3. Smart Manufacturing Monitoring 4. Heterogeneous Edge Acceleration.

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2)]	Start of the Project	Level (3)	End of the Project	Level (5)

R&D Objectives

The proposed R&D advances our foundation-model virtual sensing (FM-VS) capability [5,6] into an energy-efficient, on-site edge monitoring module for measurement-limited dynamic systems in smart manufacturing and advanced nuclear environments [4]. This effort is driven by the requirement for measurement-equivalent monitoring in advanced manufacturing lines, thermal-fluid processes, and next-generation nuclear assets where direct instrumentation is constrained by power, access, cost, and connectivity limits, exposing the scalability limitations of case-specific virtual sensing approaches (Fig. 3-1). The project transitions validated operator-learning prototypes [6,7,8] into three deployment-ready deliverables: a reusable FM-VS operator core adaptable across industrial configurations, a structured integration workflow with defined acceptance criteria aligned to SME product requirements, and a hardware-validated deployability evidence package reported in field-relevant units including measured energy per inference and tail latency under realistic industrial operating constraints.

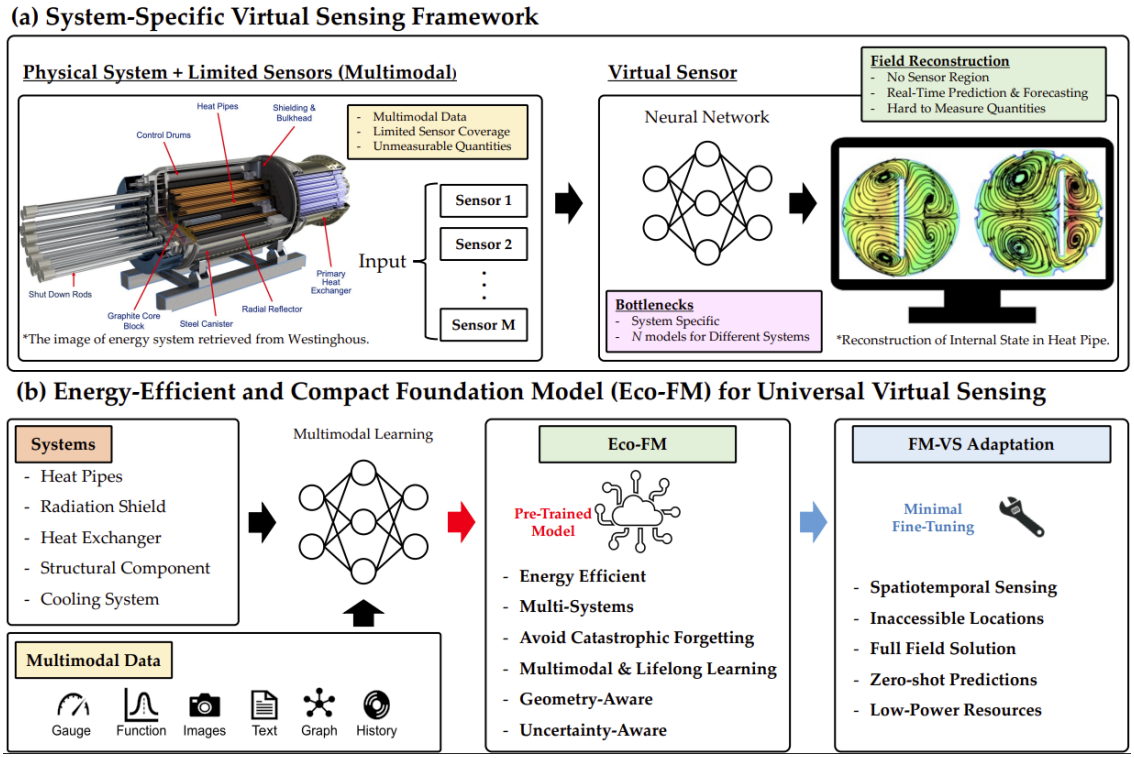

Figure 3-1. When virtual sensing is developed case by case, changes in the target asset, sensor layout, or operating regime trigger redevelopment and revalidation, whereas a reusable (FM-VS) approach reuses a common core and adapts to new cases under a consistent evaluation protocol. [6,9,12]

Technologies currently available in our group

Our group has established the core technical basis for foundation-model-enabled virtual sensing in physical systems:

- *A compact foundation model (Fig. 3-2) based on neural-operator building blocks for reconstructing full-field states from sparse and asynchronous sensing [10,11,14].*
- *Neural-inspired architectures evaluated on nuclear cases with potential for improved energy-efficiency within neuromorphic hardware implementations (Fig. 3-3) [9]*
- *Geometry-aware extensions designed to support transfer across component geometries and operating conditions [15,16].*

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- *A deployment-driven evaluation framing that treats energy and latency for edge execution as first-class metrics, alongside reconstruction quality and robustness.*
- *In-house capability: model development, edge execution, and accelerator-oriented profiling to inform heterogeneous deployment under power and latency constraints.*

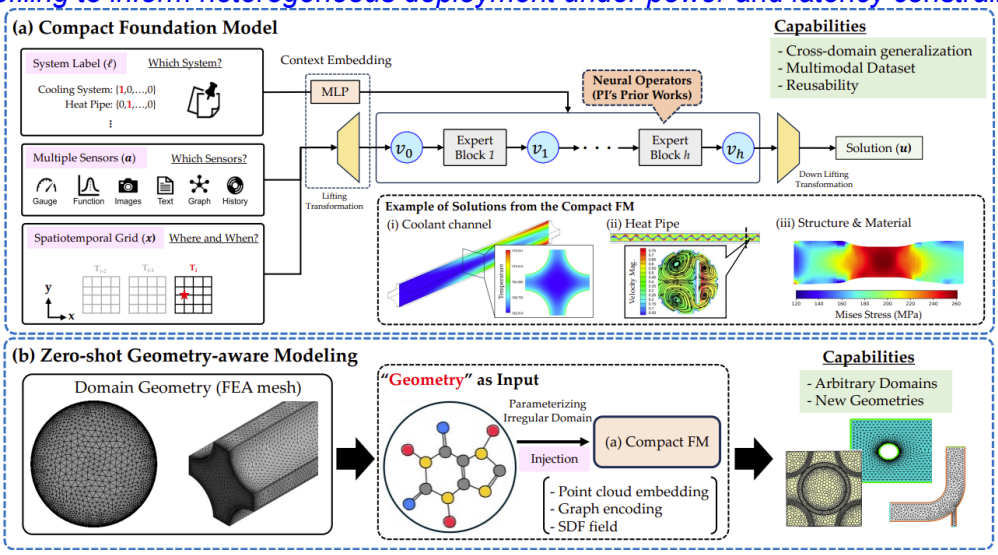

Figure 3-2. The FM-VS core reconstructs full-field states from sparse measurements using an operator-based backbone with conditioning across contexts and geometries, and the surrounding deployability layer frames edge feasibility by reporting energy per inference and tail latency together with reconstruction quality while supporting practical execution paths such as quantized inference on representative edge platforms [6,9,12,15,16].

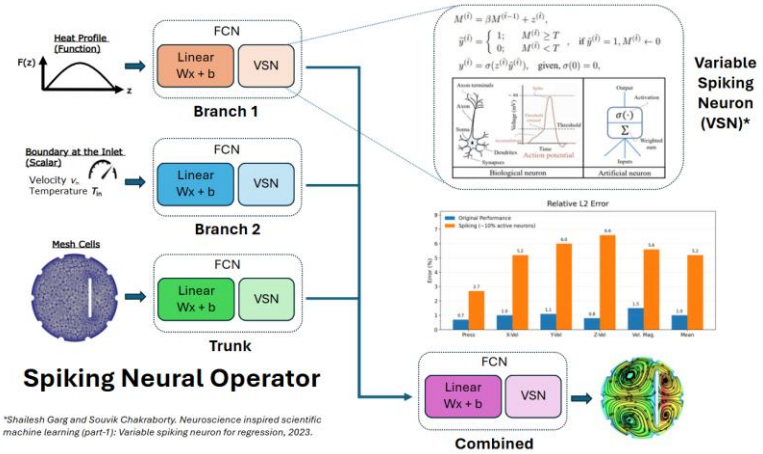

Figure 3-3. This spiking-operator architecture maps boundary conditions and sparse measurements through a branch–trunk structure to a reconstructed field, and it enables event-driven updates through spiking dynamics as an optional low-power pathway that can be evaluated using the same deployability metrics of energy per inference and tail latency [9].

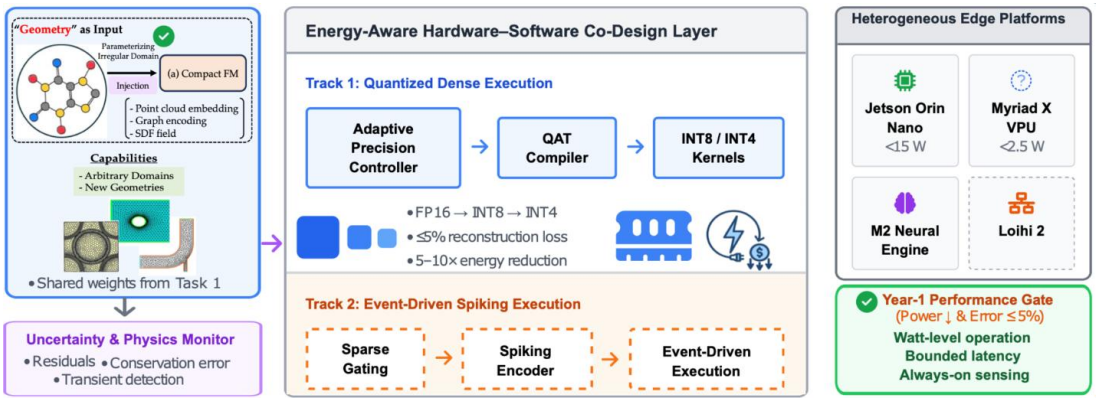

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Figure 3-4. Energy-efficient FM with edge execution. The FM is deployed via hardware–software co-design for watt-level operation.

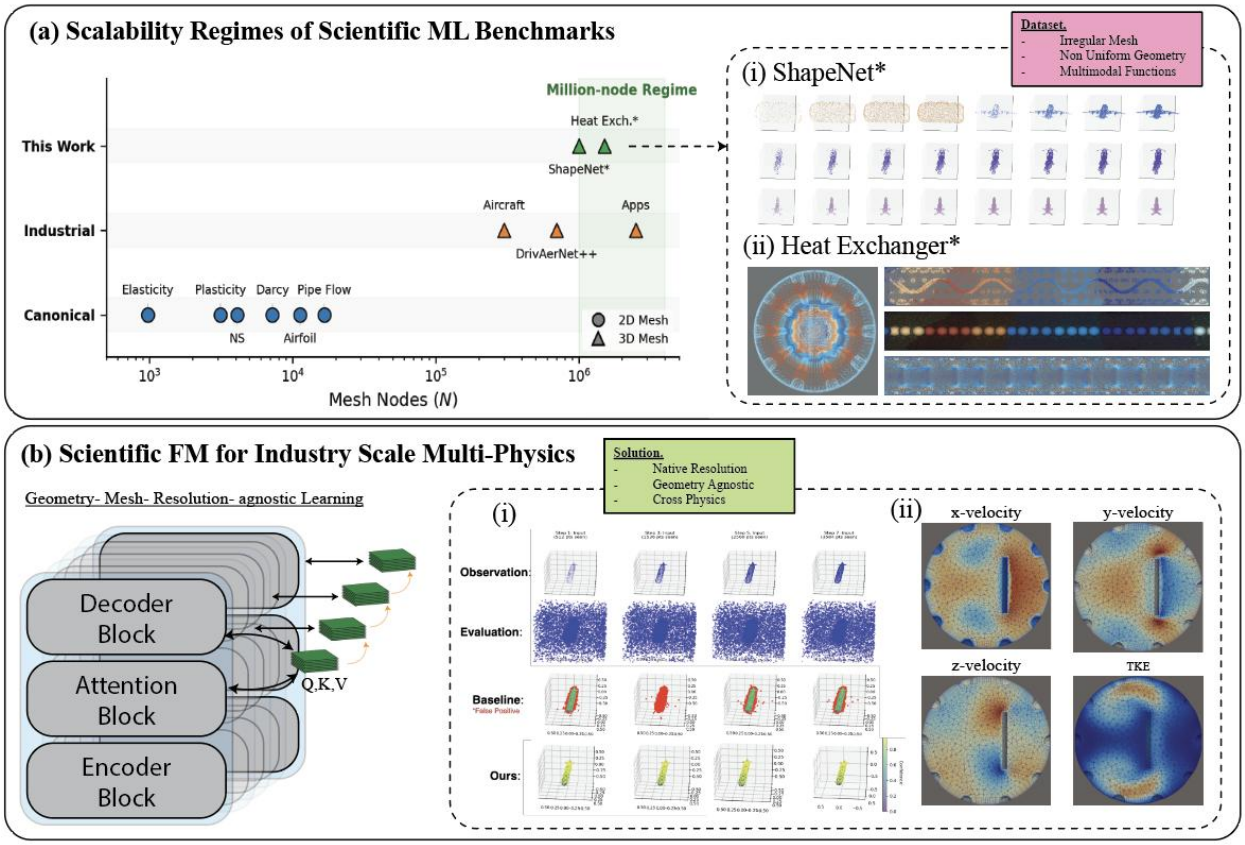

Figure 3-5. Industry-scale physics solvers require capacity to handle over-million nodes of scientific data. However, this requirement is not met for general ML driven models due to over-quadratic memory requirement growth in respect to the datapoint, as well as the well-known curse of dimensionality [17]. We propose Geometry- Mesh- Resolution Agnostic Scientific Foundation Model for Industrial Scale Cross Physics Learning using Attention Operator Architecture [11] with capacity to handle over-million nodes of irregular mesh.

Technologies to be developed further through this joint R&D

The joint R&D will mature the above capabilities into SME-usable deliverables by strengthening deployability evidence and integration-facing outputs:

- *A reusable FM-VS core and a partner-facing integration workflow, with clear interfaces and acceptance criteria suitable for adoption across multiple asset classes.*
- *Further sophisticated design for spiking architectures within our FM-VS framework, along with neuromorphic hardware implementation, potentially utilizing existing technologies such as Loihi 2 for concrete power reduction and presentable statistics*
- *Quantization-aware training of FM-VS framework with limited degradation for a targeted 5-10x energy reduction utilizing adaptive precision driven by uncertainty and duty-cycled sampling (Fig. 3-4).*
- *Hardware-validated deployability evidence on representative edge platforms, reported as energy per inference and tail latency to support system-level tradeoffs.*
- *A hardware-software co-design and benchmarking pathway for heterogeneous edge execution, including accelerator-class options (for example, FPGA-class) and exploratory neuromorphic-class evaluation where appropriate for continuous, power-constrained operation.*

R&D Target Specifications

Target specifications will be demonstrated on representative problems and reported in accuracy, energy/inference, and latency to support evaluation for on-site edge deployment.

- ***Throughput vs simulation:*** *~1000× speedup compared to high-fidelity FEA on representative cases (see Fig. 3-5).*

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- ***Quantized deployment quality:*** *≤ 5% reconstruction degradation relative to an FP16 baseline under the quantized deployment path.*
- ***Quantized deployment energy:*** *5–10× energy reduction at batch-1 inference under the quantized deployment path.*
- ***On-site operating envelope:*** *< 1 Hz steady-state monitoring and 10 Hz during transients using duty-cycled sampling.*
- ***Benchmark hardware context:*** *Jetson Orin Nano under 15 W, Myriad X VPU under 2.5 W, and the M2 Neural Engine for quantized dense benchmarking.*
- ***Trustworthiness:*** *≥90% distribution-free coverage for uncertainty outputs.*
- ***Year-1 gate:*** *sustained operation under strict power and timing limits with error <5% as the go/no-go criterion.*

R&D Plan

The project translates our validated foundation-model capability into an integration-ready module suitable for field evaluation by a Korean SME. The work delivers a reusable and deployment-qualified FM-VS operator core, expressed in edge-relevant units including measured energy per inference and tail latency, together with a structured integration workflow and validation package aligned with defined acceptance criteria (Fig. 3-6).

To anchor the architecture in a controlled physical environment, we are developing the Martian Lab, a two-tank coupled-dynamics testbed at UIUC NPRE that serves as the project's end-to-end validation platform. The testbed integrates level, flow, and fiber-optic sensing with motor-operated valves and pump actuation, replicating the sensor-sparse and actuator-coupled dynamics characteristic of smart manufacturing and industrial energy systems. Fig. 3-7 presents the full data pipeline, separating a training stage in which sensor streams and high-fidelity simulation data jointly inform operator learning, from an inference stage in which the trained model is deployed on energy-efficient edge hardware to provide real-time actuator feedback, virtual sensing of unmeasured states, and predictive maintenance signals through a cloud-connected digital twin [12,13,17].

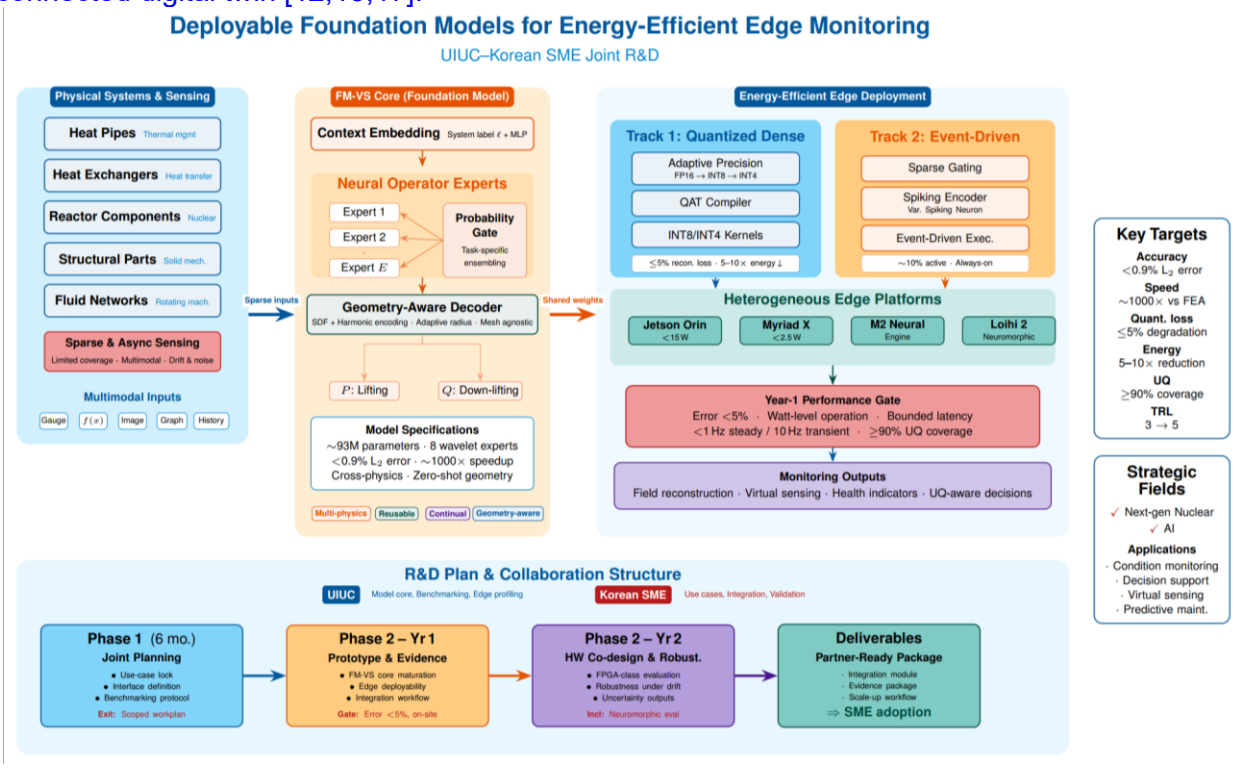

Figure 3-6. System overview of the deployable physics foundation model (FM-VS) and energy-efficient edge deployment tracks with validation gates. *Sparse, asynchronous inputs from measurement-limited energy assets (heat pipes, heat exchangers, reactor components, fluid networks) are processed by the FM-VS core, where neural operator experts with task-specific probability gating and a geometry-aware decoder produce full-field reconstructions from a compact 93M-parameter model (~1000× speedup over FEA). The trained model is compressed along two parallel edge tracks—quantized dense (FP16→INT8/INT4, ≤5% reconstruction loss, 5–10× energy reduction) and*

event-driven (spiking encoders, ≥10% active compute): for deployment on heterogeneous hardware (Jetson Orin ≤15 W, Myriad X ≤2.5 W, M2 Neural Engine, Loihi 2). A Year-1 performance gate enforces error <5%, watt-level operation, bounded latency (≤1 Hz steady/10 Hz transient), and ≥90% UQ coverage before field handoff. Monitoring outputs: field reconstruction, virtual sensing, health indicators, and UQ-aware decisions--address next-generation nuclear and industrial applications. The phased R&D collaboration assigns UIUC the model core, benchmarking, and edge profiling, while the Korean SME contributes use cases, integration, and validation, culminating in a partner-ready deliverables package for SME adoption.

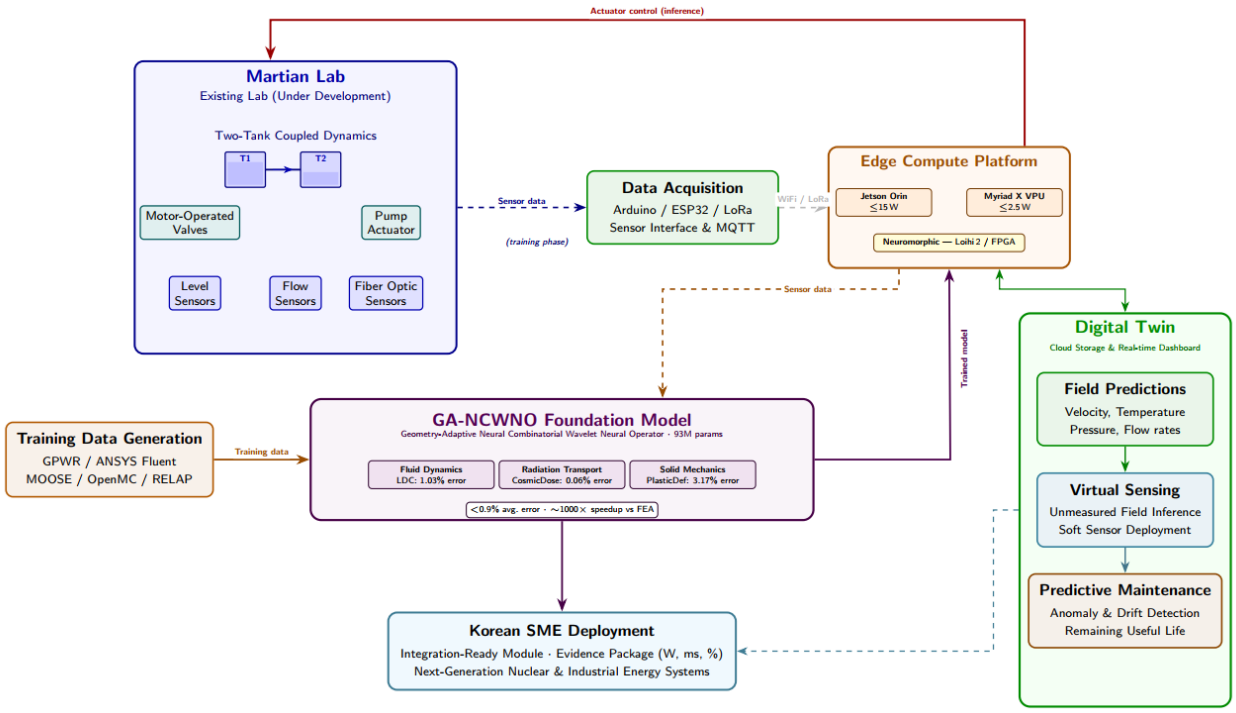

***Figure 3-7**. System architecture of the Martian Lab deployable physics foundation model framework. During the training phase (dashed arrows), sensor data from the two-tank coupled dynamics testbed is collected via Arduino/ESP32 data acquisition and transmitted to the Edge Compute Platform, where it supplements high-fidelity simulation data (GPWR, ANSYS Fluent etc.) to train the foundation model. The trained 93M-parameter model achieves <0.9% average error across fluid dynamics, radiation transport, and solid mechanics with ~1000× speedup over FEA. During the inference phase (solid arrows), the trained model is deployed on energy-efficient edge hardware (Jetson Orin ≤15W, Myriad X VPU ≤2.5W, Neuromorphic Loihi 2/FPGA) to provide real-time actuator control to the Martian Lab and feed a cloud-based Digital Twin for field predictions, virtual sensing, and predictive maintenance. The integration-ready module with evidence package is delivered to Korean SME partners for deployment in next-generation nuclear and industrial energy systems.*

The two-stage architecture in Figure 3-7 directly maps onto the phased R&D structure. Phase 1 formalizes data interfaces, acceptance thresholds, and benchmarking protocols under defined power and latency constraints. Phase 2 advances each component from laboratory validation to hardware-qualified, integration-ready deployment.

Phase 1 (6 months): Joint planning and de-risking (Seed stage)

1) **Use-case and requirements lock.** Select representative asset class(es) and sensing configurations, and define acceptance criteria for reconstruction quality, robustness, and edge deployability under on-site constraints.
2) **Data and interface definition.** Specify measurement formats, timing (sparse and asynchronous inputs), and interfaces required to connect the model to the partner's monitoring workflow.
3) **Initial benchmarking protocol.** Establish a shared evaluation protocol that defines the reporting package: reconstruction metrics plus measured energy per inference and tail latency on representative edge platforms.

Phase 1 exit criteria. A jointly scoped Phase 2 workplan, a finalized evaluation protocol, and an agreed acceptance checklist suitable for Phase 2 submission.

Phase 2 (2.5 years): Joint development and validation

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	Year 1: Integration-ready prototype and deployability evidence. 1) **Reusable FM-VS core maturation.** Strengthen reuse across operating regimes and component geometries and define a stable model interface for integration. 2) **Energy-Efficient FM: Hardware-Software Implementation and Integration** a. **Sophisticated Neuromorphic Sensing.** Further our development of neural-inspired virtual sensing architectures within FM-VS framework along with initial neuromorphic hardware implementation (e.g. Loihi 2) b. **Quantization-based Efficiency.** Implement adaptive precision methodologies to eventual FM-VS for representative edge platforms (e.g. Jetson Orin Nano) 3) **Edge deployability validation.** Produce hardware-validated measurements of energy per inference and tail latency, reported alongside reconstruction quality and robustness, using the shared protocol. 4) **Initial integration workflow.** Deliver a partner-facing workflow that specifies inputs, outputs, deployment steps, and acceptance checks aligned with the Phase 1 protocol. Year 1 performance checkpoint. Demonstrate sustained on-site operation within defined power and timing limits while meeting the agreed accuracy and robustness threshold. Year 2: Heterogeneous execution options and robustness hardening 1) **Heterogeneous execution pathway.** Establish a profiling-driven hardware–software co-design pathway for heterogeneous edge execution. Accelerator-class options, including FPGA-class execution, are evaluated under the same reporting protocol when they provide clear power or latency advantages. 2) **Robustness and uncertainty for field operation.** Strengthen performance under sensing sparsity, drift, and operating shifts, and provide uncertainty-aware outputs from our quantization-aware and neural-inspired FM-VS core appropriate for long-lived monitoring. **Final period:** Partner-ready validation package: Partner-ready evidence package. Deliver an integration-ready module and a validation package demonstrating performance on representative scenarios agreed in Phase 1, including standardized reporting of reconstruction quality, energy per inference, and tail latency to support SME decision-making.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**Technical contributions** • **FM-VS model core**: Field reconstruction from sparse and asynchronous measurements. • **Energy-Efficient Hardware-Software Co-Design:** neural-inspired architectures within our FM-VS framework and quantization-aware methodologies along with hardware implementation • **Reuse enablers**: geometry-aware and reuse-oriented components to support transfer across operating regimes and configurations. • **Deployment evaluation:** benchmarking methodology that reports reconstruction quality together with energy per inference and tail latency. **Research scope** • Lead model development, controlled evaluation, and deployability measurement required to mature the capability into an integration-ready module. • Partner-specific product requirements, integration constraints, and validation context will be defined by the participating Korean SME and refined jointly during Phase 1, while UIUC focuses on delivering the core technology and deployability evidence. **Resources to be provided** • PI technical leadership and research staff effort for model development, benchmarking implementation, and documentation. • In-lab compute and development infrastructure for training, evaluation, and reproducible benchmarking.

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	• Engineering support for edge execution and accelerator-oriented profiling, sufficient to generate partner-facing evidence in standardized reporting format. **Specific tasks to be provided** • Phase 1a: Co-define acceptance criteria and evaluation protocol, including measurement formats, interface assumptions, and benchmark aligned with SME decision needs. • Phase 1b: Deliver an initial deployability evidence package on representative edge platforms, including measured energy/inference and tail latency under agreed conditions. • Phase 2a: Mature the reusable FM-VS core and deliver a stable integration interface with documented inputs, outputs, and acceptance checks. • Phase 2b: Implement hardware-software collaboration for improved energy-efficiency utilizing quantization-aware techniques and neuromorphic computing • Phase 2c: Produce hardware-validated deployability evidence under the shared protocol, and support heterogeneous execution benchmarking (including FPGA-class evaluation when it offers clear power or latency benefit). • Phase 2d: Deliver a partner-facing integration package and final validation evidence package suitable for SME adoption planning.
Expected R&R of Korean SME Partner	**Technical contributions** • Define the target asset class and monitoring use case, including operating modes, sensing constraints, and deployment boundaries relevant to the SME context. • Provide domain input to define acceptance criteria for monitoring outputs, including the health indicators of interest, decision thresholds, and reporting cadence. **Development and integration tasks** • Lead system integration of the FM-VS module into the SME monitoring workflow, including data acquisition, preprocessing, and interface adaptation aligned with the agreed evaluation protocol. • Implement deployment-facing software components needed in the target environment: logging, fault handling, and update procedures consistent with SME operational practice. **Testing and validation activities** • Provide representative datasets, sensing configurations, and operational scenarios including sparse and asynchronous sensing conditions and realistic drift cases. • Lead validation on SME-relevant platforms and environments, including confirmation that outputs are actionable for the intended workflow. • Participate in joint benchmark reviews and confirm acceptance outcomes against the Phase 1 checklist and Phase 2 evaluation protocol. **Resource commitments** • Assign technical staff to support Phase 1 requirements definition and interface alignment, and Phase 2 integration and validation activities. • Provide access to representative hardware, instrumentation context, and test environments needed to evaluate on-site constraints, subject to SME safety and confidentiality requirements. **Collaboration efforts** • Co-develop and maintain the evaluation protocol and acceptance checklist through regular technical reviews. • Provide timely feedback on integration constraints and deployment friction points to guide model updates and benchmarking priorities. • Lead SME-side proposal preparation and reporting for the staged TIPA process, with UIUC providing technical inputs and evidence packages as needed.

5. Expected Outcomes/Impact

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Expected Outcomes and Impact	**Expected outcomes** • **Integration-ready package.** A partner-ready software package with defined inputs and outputs, interface assumptions for sparse and asynchronous sensing, and acceptance checks aligned with the Phase 1 evaluation protocol. • **Measured deployability evidence.** A consolidated evidence package reporting reconstruction quality together with energy per inference and tail latency on representative edge platforms, with documented test conditions suitable for review. • **Scale-up workflow for partners.** A repeatable benchmarking workflow that enables SMEs to compare heterogeneous edge options using consistent deployability metrics and to plan **product-level integration across manufacturing and advanced energy applications**. **Commercialization potential** • The project is designed for commercial deployment, not just research publication. Every output includes measurable, hardware-validated evidence and clear acceptance checks that an SME can directly map to its product requirements and integration constraints. • The same capability can support multiple manufacturing and advanced energy asset classes where dense sensing is impractical and on-site operation is required, allowing partners to extend the module beyond a single use case. **Broader impact** • In safety-**and productivity-**critical infrastructure, reliably estimating internal states from limited measurements can mean the difference between detecting degradation early and discovering it after failure. This capability enables **informed maintenance decisions, improved yield,** and supports the transition from time-based to condition-based monitoring in **smart manufacturing and next-generation energy systems**. • Standardized interfaces and evidence-based reporting can lower adoption barriers for multiple SMEs and strengthen the industrial base for edge AI monitoring.

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